

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL

8223/1

PAPER 1 Multiple Choice

1 hour

Marks 40

2021

Additional Materials: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write in soft pencil.
- Make sure that you receive the multiple choice answer sheet with your examination number on it.
- There are **forty** questions on this paper. Answer **all** questions.
- For each question, there are four possible answers **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.
- If you want to change an answer, thoroughly erase the one you wish to delete.

Read the instructions on the answer sheet carefully.

- Each correct answer will score one mark.
- All questions in this paper carry equal marks.
- Any rough working should be done in this booklet.
- You may use a non-programmable calculator.

This document consists of **13** printed pages and **3** blank pages.



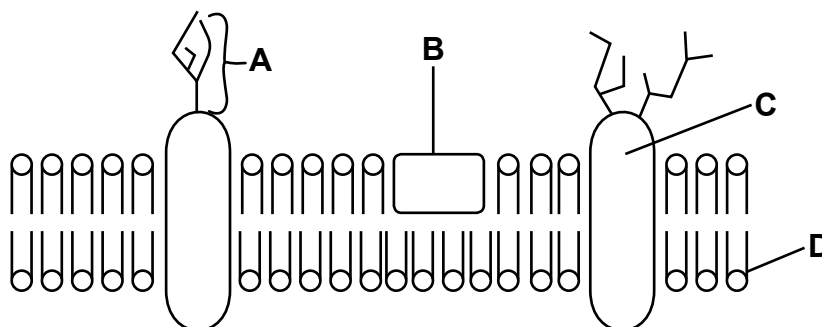
Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 Which group of organisms must be a single population?
 - A all the living organisms on an island in the middle of a river
 - B all the insects occupying a mahangu field
 - C all the birds seen on one day at a watering hole
 - D all the bacteria in a colony of *Vibrio cholerae*
- 2 Which term is used to describe all organisms in an area and their interactions with each other and with their environment?
 - A ecosystem
 - B niche
 - C habitat
 - D species

- 3 The diagram shows a cell surface membrane.

Which letter indicates a glycoprotein?

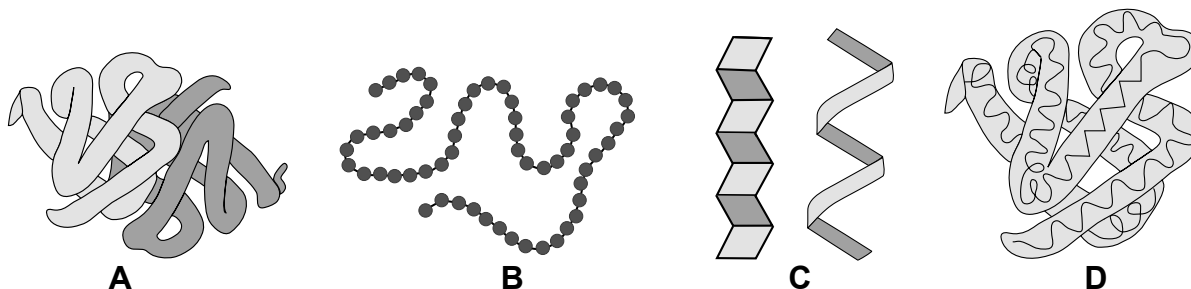


- 4 A learner wrote statements about the functions of cholesterol molecules.
 - 1 they stabilise membrane structure
 - 2 they make the membrane more fluid at low temperatures and less fluid at high temperatures
 - 3 they transport substances from one side of the membrane to the other side

Which statements are correct?

- A 1 and 2 only B 2 and 3 only C 1 only D 2 only
- 5 A root hair cell was placed in a beaker containing solution X. After 2 hours, the root hair cell was removed and was observed, using a light microscope. It was observed that its cell membrane had partly detached from the cell wall. Which statement describes solution X?
 - A has higher water potential than the root hair cell
 - B has lower water potential than the root hair cell
 - C has the same water potential as the root hair cell
 - D it is distilled water

6 Which diagram represents the tertiary structure of protein?



7 What occurs as a result of a condensation reaction?

- A breakdown of monomers into polymers
- B breaking of glycosidic bonds
- C formation of covalent bonds
- D production of water molecule

8 What are the components of an amino acid?

	amino group	carboxyl group	R side chain
A	✓	x	✓
B	x	✓	x
C	✓	x	x
D	✓	✓	✓

Key: ✓ = yes

x = no

9 Which must be true for a competitive inhibitor?

- A it must be a poison
- B it must be a highly reactive compound
- C it must be structurally similar to the substrate
- D it must be a metal ion such as Fe^{2+}

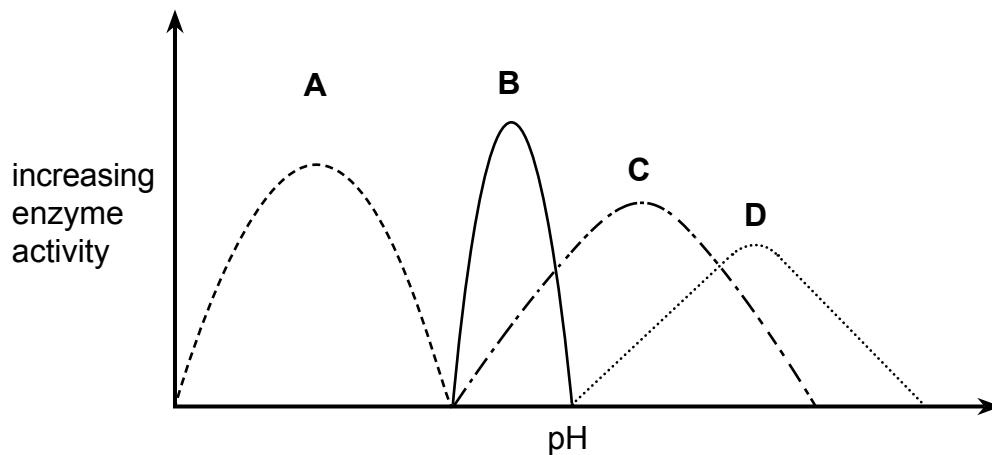
10 Which statements are true for a non-competitive inhibitor of an enzyme-catalysed reaction?

- 1 it decreases $V_{\max}$
- 2 it binds to the enzyme-substrate-complex (ES)
- 3 it does not have an effect when substrate concentrations are increased

- A** 1 only
B 2 only
C 1 and 2 only
D 1, 2 and 3

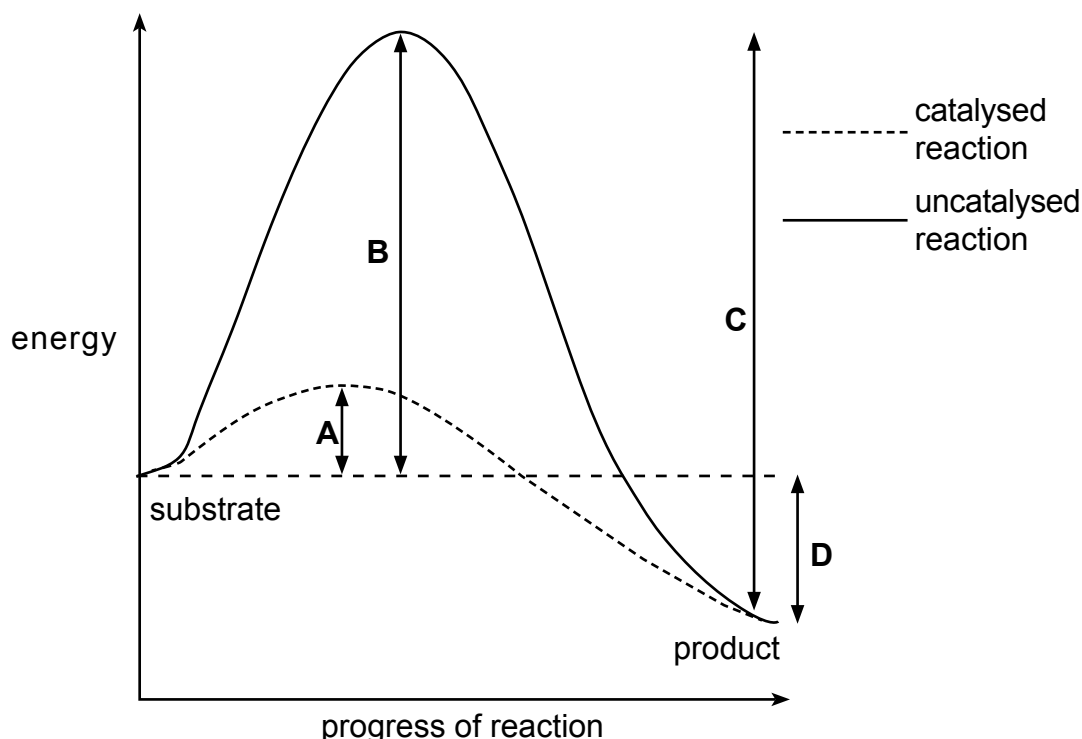
11 The graph shows the reaction rate of four enzymes **A**, **B**, **C** and **D**.

Which enzyme shows the greatest change in reaction rate over the smallest pH range?



- 12** The graph shows the progress of a reaction in the presence and absence of an enzyme.

Which letter **A**, **B**, **C** or **D** represents the activation energy of a reaction in the absence of an enzyme?



- 13** A learner completed an investigation and found that a starch-digesting enzyme has its highest activity at pH 7 and a temperature of 37°C. The learner repeated the investigation. The substrate is in excess.

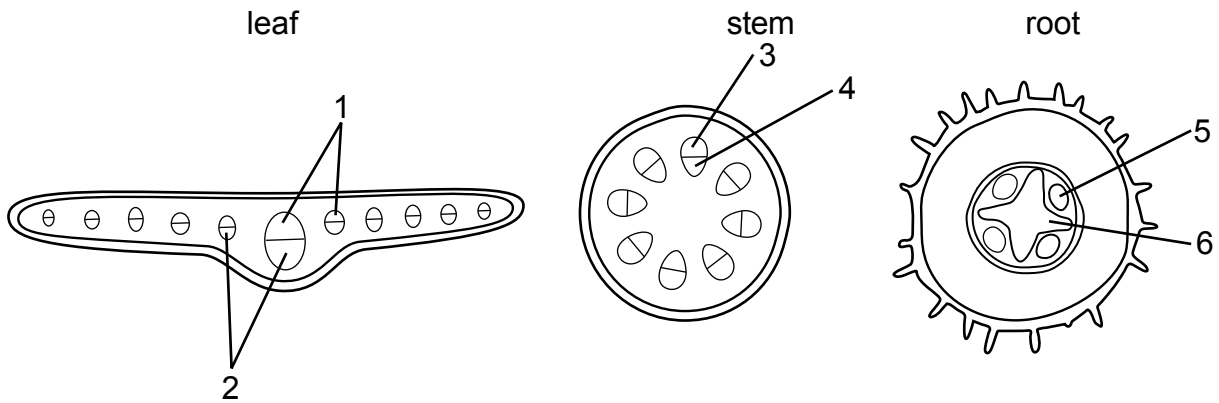
Which variable could the learner change to increase the rate of reaction?

- A** increasing the concentration of enzyme
 - B** decreasing the concentration of enzyme
 - C** increasing the temperature to 45°C
 - D** decreasing the temperature to 25°C
- 14** When water moves from roots to the leaves, it flows through cell walls, cytoplasm and plasmodesmata.

Which row correctly links the pathways through which water flows and the named structures?

	cell walls	cytoplasm	plasmodesmata
A	symplastic	apoplastic	apoplastic
B	apoplastic	symplastic	symplastic
C	symplastic	apoplastic	symplastic
D	apoplastic	symplastic	apoplastic

15 The diagrams show transverse sections through three plant organs.



Which labelled regions of tissue transport water and mineral ions?

- A** 1, 3 and 5
- B** 2, 3 and 6
- C** 1, 4 and 6
- D** 2, 4 and 5

16 Which row correctly describes the function of each named structure?

	xylem vessel elements	phloem sieve tube elements	companion cells
A	support the plant	transport carbohydrates	load and unload sugar
B	transport water and mineral ions	support the plant	allow the transport of large volumes of water
C	form a continuous tube	allow the transport of large volumes of water	support the plant
D	form a long hollow tube	form a continuous tube	transport carbohydrates

17 The wall of an artery contains elastic fibres.

Which of the statements are true about the main function of these fibres in the artery wall?

- 1 keep blood moving away from the heart
- 2 recoil to maintain a high blood pressure
- 3 help protect the artery from high pressure

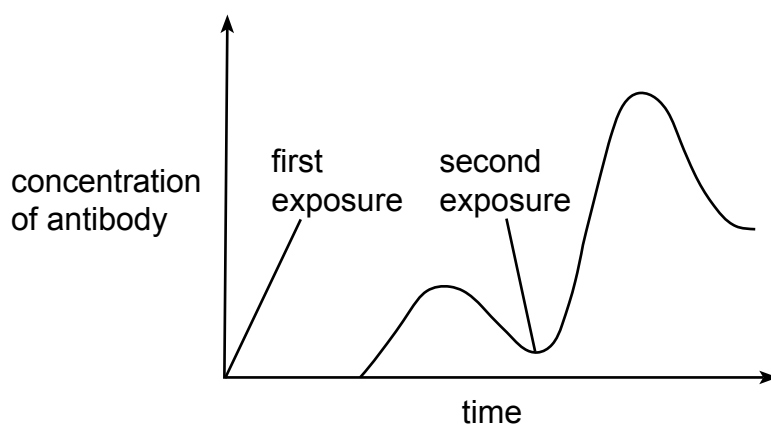
- A** 1 only
- B** 1 and 2 only
- C** 1 and 3 only
- D** 1, 2 and 3

18 Which components of blood are also present in lymph?

	white blood cells	proteins	mineral ions
A	✓	✓	✗
B	✗	✗	✗
C	✓	✓	✓
D	✗	✗	✓

Key: ✓ present
✗ absent

19 The diagram below shows immune responses after first and second exposure to a pathogen.



Which cells of the immune system are directly responsible for the rapid increase in antibody concentration after the second exposure to a pathogen?

- A plasma cells
- B B-memory cells
- C phagocytes
- D T-memory cells

20 Below are numbered events which take place during an immune response.

- 1 secretion of antibodies
- 2 development of plasma cells
- 3 recognition of non-self antigens
- 4 mitosis of B-lymphocytes

Which of the following is the correct sequence of events occurring during an immune response?

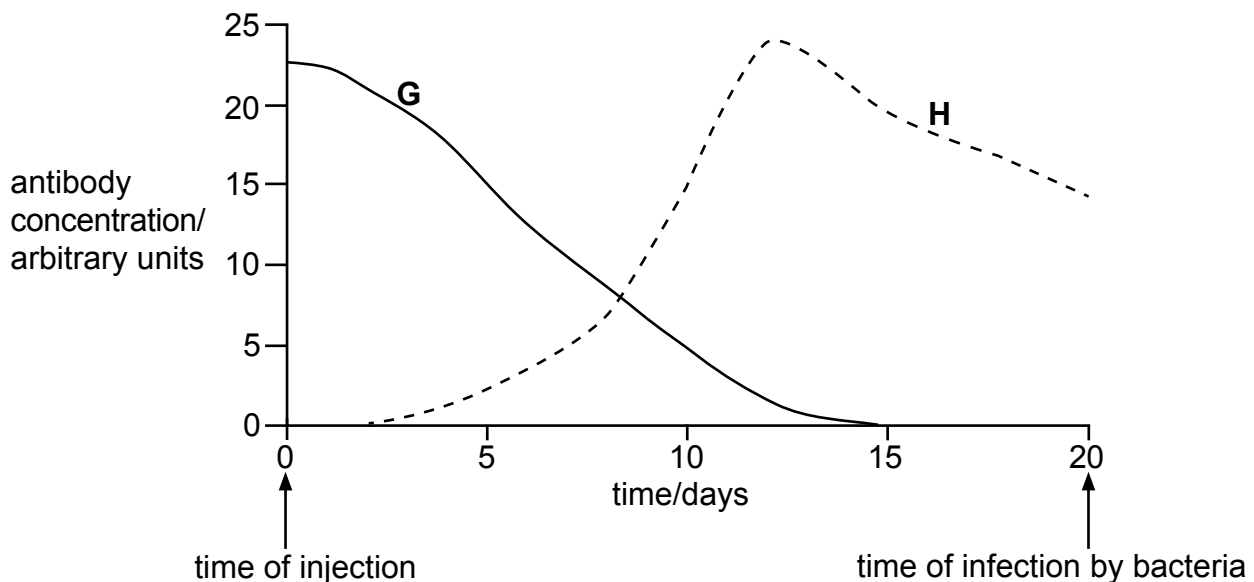
- A 2 3 4 1
- B 3 4 2 1
- C 3 1 4 2
- D 1 2 4 3

21 Which statements correctly describe the roles of lymphocytes?

- 1 some T-lymphocytes stimulate macrophages to kill infected cells
- 2 each B-lymphocyte has the ability to make several types of antibody molecules
- 3 plasma cells secrete antibodies into the blood plasma
- 4 some B-lymphocytes and T-lymphocytes become memory cells

- A** 1 and 4
B 1, 2 and 3
C 1, 3 and 4
D 2, 3 and 4

22 The graph shows the concentration of an antibody in the blood of two people, **G** and **H**, over a period of twenty days. On day zero, person **G** was injected with antibody for TB and person **H** was injected with vaccine for TB. On day twenty, persons **G** and **H** were infected with the bacteria that cause TB.



Which statement correctly describes the changes in antibody concentration after day 20?

- A** concentration of antibodies in **H** would remain constant
B antibodies of **G** increase more rapidly than antibodies for **H**
C **G** would peak in antibody production 12 days after infection and **H** will not produce any antibodies
D **H** would produce a new antibody peak 12 days after infection

- 23** Which step is **not** part of hybridoma method for production of monoclonal antibodies?
- A** antigens that cause the production of required antibodies are injected into a mammal
 - B** B-lymphocytes producing the desired antibodies against the antigen are collected
 - C** the collected B-lymphocytes producing the desired antibodies are multiplied by injecting them into a mammal
 - D** B-lymphocytes collected from the mammal are fused with B cell myeloma to produce hybridoma cells
- 24** Which tissue would be seen in a photomicrograph of the wall of the trachea?
- 1 elastic fibres
 - 2 epidermis
 - 3 smooth muscle
- A** 1, 2 and 3
 - B** 2 and 3 only
 - C** 1 and 2 only
 - D** 1 and 3 only
- 25** Which pair of parts of the respiratory tract contain cartilage?
- A** bronchus and alveolus
 - B** bronchus and trachea
 - C** bronchiole and trachea
 - D** bronchiole and alveolus
- 26** Goblet cells are found in the trachea.

Which organelles would be found in large numbers in a goblet cell?

	lysosomes	mitochondria	rough endoplasmic reticulum	secretory vesicles
A	✓	✓	✓	✓
B	✗	✓	✓	✓
C	✓	✗	✓	✗
D	✓	✓	✗	✓

Key: ✓ = present in large numbers
✗ = not present in large numbers

- 27** *Epulopiscium fishelsoni* is one of the largest bacteria which can grow up to 600 μm . The average human egg cell at ovulation is 100 000 nm. How many times is the egg cell smaller than the cell of *E. fishelsoni* ?
- A** $\times 60\,000$
B $\times 166$
C $\times 6$
D $\times 0.006$
- 28** Which factors maintain the diffusion gradient for carbon dioxide at the surface of the alveoli?
- 1 blood flow past the alveoli
 - 2 breathing movement exchanging air in the lungs
 - 3 thin epithelial lining of the alveoli
- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1 only
- 29** The squamous epithelial cells of the alveoli form part of the gas exchange system. How do these cells assist gas exchange?
- 1 they provide a large surface area
 - 2 they contain many mitochondria
 - 3 they provide a short diffusion distance
- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3
- 30** Which row is correct regarding the effect of tar on the human gas exchange system?

	mucus glands in the trachea are enlarged	mutations may occur in epithelial cells forming tumors	goblet cells are stimulated to secrete more mucus
A	✓	✓	✓
B	✓	✗	✗
C	✗	✓	✓
D	✗	✓	✗

Key: ✓ = true

✗ = false

31 Which structure is **not** present in a typical prokaryotic cell?

- A** 70S ribosome
- B** peptidoglycan cell wall
- C** plasmodesmata
- D** naked circular DNA

32 Which row correctly shows the short-term effects of nicotine and carbon monoxide in cigarette smoke?

	nicotine	carbon monoxide
A	decreases heart rate	decreases blood pressure
B	increase blood flow to the feet	increases blood pressure
C	increases blood pressure	decreases oxygen transport in blood
D	increases blood pressure	increases heart rate

33 Each statement describes part of the process of mitosis.

- 1 Chromosomes align along the equator.
- 2 Chromatids uncoil and form a chromatin network.
- 3 Centromeres split, allowing chromatids to separate.
- 4 Chromosomes condense and become visible.

Which sequence of events is correct?

- A** 2 3 4 1
- B** 4 3 2 1
- C** 4 1 3 2
- D** 1 2 4 3

34 Which stage of mitosis is shown by cell **X**?



- A** prophase
- B** metaphase
- C** anaphase
- D** telophase

35 Which statement correctly describes the characteristics of daughter cells produced during mitosis compared to the parent cell?

- A** same numbers of chromosomes with different genetic code
- B** same number of chromosomes with the same genetic code
- C** half number of chromosomes with the same genetic code
- D** half number of chromosomes with different genetic code

36 What is replicated during the S phase in the cell cycle?

- A** RNA
- B** proteins
- C** RNA and proteins
- D** DNA

37 Which row is correct about the structure of a chromosome during metaphase of mitosis?

	number of centromeres	number of chromatids	number of polynucleotide strands	number of telomeres
A	2	2	2	2
B	2	1	4	4
C	1	2	2	4
D	1	2	4	4

38 Which of the following is **not** part of a nucleotide?

- A** phosphate group
- B** pentose sugar
- C** nitrogenous base
- D** polynucleotide

39 Which of the following bases are purines?

- 1 adenine
- 2 thymine
- 3 guanine
- 4 cytosine

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 2 and 4 only

- 40** How many hydrogen bonds are found between guanine and cytosine in a double stranded DNA molecule?
- A** 1
 - B** 2
 - C** 3
 - D** 4

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8223/1/21